

David F. Akinremi

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EDUCATION

Tufts University

Bachelor of Science in Computer Science; Minor in Entrepreneurship

Medford, MA

September 2020 – May 2024

Achievements:

- Dean's List for six semesters at Tufts University.
- Finalist in the **NASA MITTIC** (MUREP Innovation and Tech Transfer Idea Competition) in 2022 and presented a more efficient and sustainable battery model to power electric vehicles and sports cars.

Relevant Coursework: Data Structures, Algorithms, Web Programming, Machine Structure and Assembly Language, Intro to Security, Computation Theory, Game Design, Calculus II.

SKILLS

Languages: C++, C#, C, Python, HTML, CSS, Javascript, XML.

Technologies: React.js, Tailwind CSS, Figma, Framer, Git/Github, Canva, Unity, Blender, Photoshop.

EXPERIENCE

Weather Research & Forecasting (WRF) – Hampton University

Hampton, VA

Software Engineer Researcher | Atmospheric Science Team

June 2023 – July 2023

- Accomplished an **8-week** research project within the **NASA-funded CURE** (Course-based Undergrad Research Experiences) program at Hampton University, focusing on enhancing the accuracy of regional weather forecasts.
- Debugged a **10-year-old legacy** implementation of the WRF model. It had been unable to build/run remotely in the last decade.
- Modernized the dependency management using a config system and successfully built and deployed the model on a remote server.

PROJECTS

Todo List Web App | React.js, Tailwind CSS

May 2024 - June 2024

- Developed a dynamic Todo List application that enhances daily productivity and motivation.
- Implemented a responsive front-end for a web application using React.js, styled with Tailwind CSS, and enhanced with API integration for real-time data retrieval, ensuring a seamless and personalized user experience.
- Successfully integrated key features such as task management (add, edit, save, delete) and daily inspirational quotes fetched from a **REST API**, all within a user-friendly interface, enhancing user engagement and productivity.

Image Compressor/Decompressor (Arithmetic Coding) | In C language

September 2022 - October 2022

- Engineered a sophisticated image compression and decompression system to optimize storage efficiency and maintain high fidelity during image processing.
- Developing a solution that applies arithmetic coding techniques capable of reducing image file sizes significantly while preserving original image quality using **C**.
- Implemented a compression system by encoding RGB data into chrominance and luminance components using Discrete Cosine Transform (DCT) and bit-packing techniques, and developed a corresponding decompression algorithm to reconstruct images, ensuring data integrity and efficiency through structured block processing and error management.

Sudoku Solution Checker | In C language

September 2022 - October 2022

- Developed a robust program to validate the accuracy of Sudoku solutions, enhancing the reliability of solution checks.
- Creating an algorithm capable of confirming the validity of a given Sudoku grid by ensuring all rows, columns, and 3x3 subgrids contain unique numbers from 1 to 9 without any repetitions.
- Utilized **C** to implement efficient data structures, including 2D arrays for grid storage and manipulation. Engineered file I/O operations to read Sudoku solutions in raster format.
- Applied advanced computational logic to perform comprehensive checks across rows, columns, and subgrids, returning a Boolean value indicating the correctness of the solution.

Cummings Crosswalk 2D Game | C#, Unity

January 2023 – May 2023

- Developed a multi-leveled 2D Road-Crossing game using Unity and C#, to enhance strategic thinking and reaction skills through a challenging virtual environment.
- Integrated complex game logic and collision detection with dynamically generated traffic patterns and custom graphics and sound effects to simulate a real-world crossroad scenario at Tufts University, significantly boosting player engagement.

E-commerce Website | HTML, CSS, Javascript, MongoDB

January 2023 – May 2023

- Developed a fully responsive e-commerce platform tailored for the sale of AI-designed hoodies, aimed at enhancing user engagement and boosting sales conversions.
- Successfully integrating advanced functionalities including a dynamic product database, sophisticated search filters, and an efficient checkout process, leading to improved customer satisfaction and operational efficiency.
- Leveraged HTML, CSS, and JavaScript to build a user-friendly frontend, utilized MongoDB for robust data management, and engineered essential features such as a contact page, review handling, and returns management to streamline user interactions and transactions.

LEADERSHIP AND INVOLVEMENTS

Organizations/Leadership: National Society of Black Engineers (*member*), Tufts Onyx Magazine (*web designer/member*), Tufts African Dance Team – COCOA (*member*), Tufts Pick-up Soccer Club (*member*).